



Calibrating Medical Vision-Language Reasoning Under Image Corruption

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Abstract

Vision-language models (VLMs) are now widely used to answer questions about medical images. A known problem is that they often sound confident even when the image does not support their answer. Earlier benchmarks test whether accuracy drops when the image is blank or wrong, but they miss something just as important: does the model’s confidence also fall when the visual evidence is gone? Put simply, is a VLM properly unsure when it cannot really see the image? To find out, we test three locally run VLMs, including a medically specialized one, on two medical VQA datasets, VQA-RAD and PathVQA, under three image conditions: intact, blank, and mismatched. Alongside accuracy, we report calibration (expected calibration error and reliability curves) and how often the answer stays the same when the image changes. The results are worrying. On VQA-RAD, removing or replacing the image lowers accuracy while mean confidence hardly moves and can even rise, so it nearly doubles the calibration error (ECE $0.23 \rightarrow 0.43$). About half of all high-confidence answers become wrong, and 48% of answers do not change at all when the image is swapped. The effect is even stronger on PathVQA, and, strikingly, it is worst for the medically specialized model, which is the most overconfident of the three. Most importantly, the models notice a blank image more easily than a believable but wrong one, which is the more dangerous case in the clinic. We release our evaluation harness so others can run the same calibration checks on medical VLMs.

Code: https://github.com/koushikchandra/clear_vlm.

1 Introduction

Vision-language models (VLMs) can now answer open questions about radiology and pathology images, and they do so in smooth, doctor-like language. Because of this, they are increasingly proposed as tools to help with diagnosis. In the clinic, however, how a model shows its uncertainty matters just as much as how often it is right. A wrong answer given with high confidence is far more dangerous than one marked as unsure, because a clinician is more likely to trust it and act on it.

Today’s hallucination benchmarks for medical VQA already test whether a model leans on the image, usually by feeding it a blank or a wrong picture. But they score only accuracy and refusal Wu et al. [2024], Gu et al. [2026]. As a result, a basic safety question stays open: when a VLM cannot actually see the right image, does its stated confidence drop, or does it stay high? We answer this question directly. Instead of accuracy alone, we measure calibration, which is simply the agreement between how confident a model is and how often it is correct, and we watch how calibration changes as we take the visual evidence away.

Central Hypothesis

A trustworthy medical VLM should grow *less* confident when the image it needs is blank, mismatched, or corrupted. We hypothesize that current models do the opposite: confidence stays high even when the visual evidence is gone, so accuracy falls but confidence does not, a calibration failure that accuracy-only benchmarks cannot see.

To keep the study focused, we organize it around five research questions:

- **RQ1 (evidence removal).** When the image is removed or replaced, does the model’s confidence fall as it should?
- **RQ2 (blank vs. wrong).** Does a model react differently to a blank image than to a believable but wrong one?
- **RQ3 (source).** Where does the miscalibration come from: closed (yes/no) questions or open ones?
- **RQ4 (severity).** Does confidence track a gradual loss of image quality, not just its sudden removal?
- **RQ5 (verbalized).** Is the confidence a clinician actually reads, the number the model states in words, any better than its internal one?

We answer all five on three locally run VLMs from different families (including a medically specialized one) and two datasets (VQA-RAD, radiology; PathVQA, pathology), under intact, blank, and mismatched images. For each setting we report accuracy with expected calibration error (ECE), reliability diagrams, the high-confidence error rate, and image-drop consistency. We release the harness so these audits are easy to repeat.

2 Related Work

Hallucination benchmarks for medical VQA. HALT-MedVQA Wu et al. [2024] turns VQA-RAD, PathVQA, and PMC-VQA into multiple-choice questions with none-of-the-above and mismatched-image options, so it can test whether models rely on the image or on training priors. MedVH Gu et al. [2026] offers a broad hallucination test that includes a “Wrongful Image” task. On the detection side, semantic-entropy methods such as VASE Liao et al. [2025] and its expanded follow-up UniVRSE Liao et al. [2026] try to catch unreliable answers. The benchmarks emphasize task accuracy and refusal, while the detection methods rank answers by estimated uncertainty; neither evaluates whether a model’s reported confidence remains calibrated as visual evidence is removed. That is the gap we fill.

Chain-of-thought reasoning faithfulness. A second line of work asks whether a medical VLM’s written reasoning is truly grounded in the image. For example, Moll et al. [2025] separates reliance on text priors from real spatial grounding, and S-Chain Le-Duc et al. [2025] adds supervision for structured visual reasoning. These papers ask whether the explanation matches the image. We ask a different but related question: does the confidence attached to an answer track how much visual evidence is available? This signal is separate from grounding faithfulness and much cheaper to measure.

Confidence calibration. Calibration has long been studied for deep networks: expected calibration error and reliability diagrams were popularized by Guo et al. Guo et al. [2017], and later work showed that language models can state their uncertainty in words Lin et al. [2022] and that semantic entropy captures uncertainty in free-form generation Kuhn et al. [2023]. In the medical setting, recent work reports that VLMs are often miscalibrated and overconfident Du et al. [2025], Senoglu

et al. [2026], and that task performance and image-quality judgments can be unstable under image corruption Ouaari et al. [2026], Cheng et al. [2025]; see Kalpelbe et al. [2025] for a survey. Our study differs in two ways. Whereas Ouaari et al. [2026] uses generic corruptions, we use a controlled removal of evidence, so a blank image and a mismatched one separate a model’s reaction to a missing image from its reaction to a wrong one. And whereas Senoglu et al. [2026] studies verbalized uncertainty alone, we show the same overconfidence in the internal token probability and stratify it by question type, which reveals that the well-calibrated closed questions are exactly the ones that collapse into confident image-independent guessing.

3 Method

Datasets. We use the test splits of VQA-RAD Lau et al. [2018] ($n=451$, radiology) and PathVQA He et al. [2021] ($n=6719$, pathology). Both mix closed (yes/no) questions with open questions about clinical images. We load the Hugging Face mirrors identified in the bibliography. Their cards label VQA-RAD as CC0-1.0 and PathVQA as MIT; both expose only image, question, and answer fields, and we use no patient identifiers or private clinical records.

Image conditions. For every (image, question) pair we query the model three times. First, we use the **intact** image, the real one. Second, we use a **blank** image, a plain gray picture of the same size that carries no visual evidence. Third, we use a **mismatched** image, a different picture drawn at random from the same dataset, which looks believable but is clinically irrelevant. A faithful model should become less confident in the second and third cases.

Confidence and metrics. We decode each answer greedily and set its confidence to the mean per-token probability of the generated text, which is the model’s own softmax confidence. We then report four numbers. Accuracy is a normalized match to the gold answer, with some leniency for yes/no. Expected calibration error (ECE) Guo et al. [2017] uses 10 equal-width bins and is shown together with reliability diagrams. The high-confidence error rate is the fraction of answers with confidence above 0.8 that turn out wrong. Finally, image-drop consistency is the fraction of answers that stay exactly the same when the image is removed, which directly measures how often the model ignores what it sees.

Models. We evaluate three locally run VLMs from different model families: two general-purpose 7B models, Qwen2.5-VL-7B-Instruct Bai et al. [2025] and LLaVA-1.6-Mistral-7B Liu et al. [2024], and one medically specialized 4B model, MedGemma-4b-it Sellergren et al. [2025]. All use the same decoding settings (greedy, at most 32 new tokens), and we run every question in both test splits, including the severity sweep (RQ4) and the verbalized-confidence study (RQ5). The Qwen and LLaVA checkpoints are Apache-2.0; MedGemma requires acceptance of Google’s Health AI Developer Foundations terms. Exact checkpoint IDs appear in the bibliography.

4 Results

4.1 RQ1–RQ2: Confidence when the image is removed or replaced

Table 1 reports all three models under the three image conditions.

Answer. (RQ1) No. Across all three models and both datasets, blanking or replacing the image lowers accuracy, but confidence does not fall by a matching amount, so calibration error grows. For Qwen2.5-VL, VQA-RAD ECE rises from 0.229 to 0.427 under blanking, and PathVQA from 0.481 to 0.576 under mismatch. The medically specialized MedGemma is the worst calibrated: on PathVQA, blanking the image cuts accuracy in half (from 0.293 to 0.139) while confidence actually rises (0.821 $\rightarrow$ 0.830), pushing ECE to 0.691 and making 82% of high-confidence answers wrong. Medical pre-training does not fix the problem; if anything, it makes the model surer of itself.

Answer. (RQ2) Yes. A blank image is noticed more than a wrong one. Under a mismatched image, every model keeps its original answer far more often than under a blank image (for example, Qwen on VQA-RAD keeps 0.483 of its answers under mismatch but only 0.157 under blank; MedGemma keeps 0.226 under mismatch versus 0.104 under blank). The confidence values tell a similar but not

Table 1: Three locally run VLMs on VQA-RAD ($n=451$) and PathVQA ($n=6719$) under three image conditions. Removing or replacing the image lowers accuracy, yet confidence stays high and calibration error (ECE) goes up: every model stays sure of itself exactly when it can see the least. The medically specialized model (MedGemma) is the most overconfident of the three. *Consist.* is the fraction of answers identical to the intact-image answer.

Model	Dataset	Condition	Acc $\uparrow$	Mean conf.	ECE $\downarrow$	Hi-conf. err. $\downarrow$	Consist.
Qwen2.5-VL-7B	VQA-RAD	Intact	0.554	0.783	0.229	0.308	–
		Blank	0.386	0.813	0.427	0.468	0.157
		Mismatch	0.439	0.806	0.367	0.463	0.483
	PathVQA	Intact	0.338	0.819	0.481	0.523	–
		Blank	0.246	0.758	0.513	0.537	0.315
		Mismatch	0.250	0.826	0.576	0.643	0.347
LLaVA-1.6-7B	VQA-RAD	Intact	0.439	0.815	0.375	0.449	–
		Blank	0.328	0.787	0.459	0.527	0.366
		Mismatch	0.353	0.813	0.461	0.545	0.528
	PathVQA	Intact	0.312	0.796	0.484	0.564	–
		Blank	0.224	0.746	0.521	0.599	0.193
		Mismatch	0.270	0.794	0.524	0.626	0.360
MedGemma-4B	VQA-RAD	Intact	0.548	0.899	0.351	0.418	–
		Blank	0.231	0.847	0.617	0.677	0.104
		Mismatch	0.337	0.886	0.550	0.653	0.226
	PathVQA	Intact	0.293	0.821	0.527	0.590	–
		Blank	0.139	0.830	0.691	0.823	0.078
		Mismatch	0.173	0.815	0.643	0.766	0.093

uniform story: confidence dips on the blank image in three of the four 7B model/dataset cells (Qwen on PathVQA drops from 0.819 to 0.758), though Qwen on VQA-RAD is an exception where it rises (0.783 to 0.813); under a mismatched image it stays at or above the intact level. In other words, the models detect the absence of an image more easily than the presence of a wrong one, which is the more dangerous case in the clinic.

Figure 1 makes the same point visually. The intact curve sits closest to the diagonal, while the blank and mismatched curves bow further below it, especially on PathVQA and for MedGemma.

4.2 RQ3: Where does the miscalibration come from?

To see where the overconfidence lives, we split the questions into closed (yes/no) and open groups (Table 2).

Answer. (RQ3) It comes from the closed questions, the ones that otherwise look most reliable (Table 2). With the intact image, closed questions are the best-calibrated group, especially for Qwen2.5-VL (VQA-RAD ECE 0.05). But once we remove or replace the image, closed-question ECE rises sharply for every model (on VQA-RAD, Qwen $0.05 \rightarrow 0.21$ and MedGemma $0.23 \rightarrow 0.43$ from intact to mismatch, with the same trend on PathVQA). A bootstrap confirms this is not just small-sample noise: for Qwen’s closed questions the intact-to-mismatch ECE increase is $+0.165$ (95% CI $[+0.083, +0.211]$). The two 7B models also fall back on a fixed, image-independent answer: under a mismatched image they repeat their original yes/no answer 58 to 65% of the time. MedGemma differs in one respect: it changes its answer far more often (closed consistency only 0.18 to 0.28), yet it stays just as overconfident, so its calibration is even worse. Open questions behave differently: they are hard and overconfident in every condition (on PathVQA, open-question accuracy is only 0.04 with ECE above 0.70 even on the correct image), so their poor calibration comes from difficulty, not from missing evidence. The dangerous pattern of confident answers that ignore the image is thus sharpest exactly where the models otherwise look most reliable.

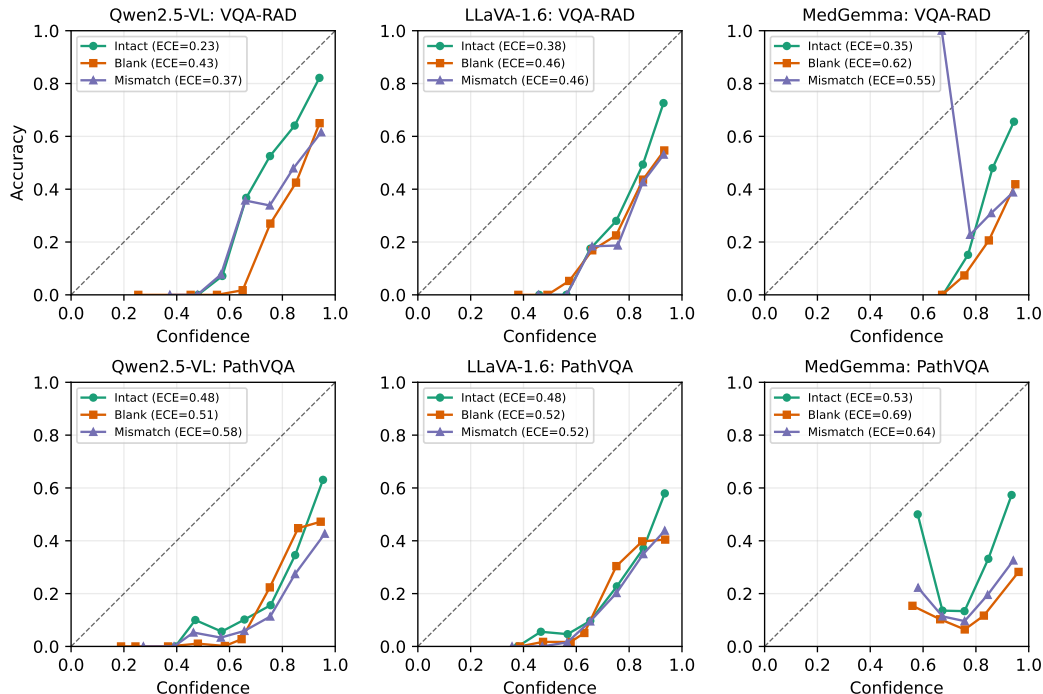


Figure 1: Reliability diagrams for all three models (columns) on both datasets (rows: VQA-RAD top, PathVQA bottom). Points on the dashed diagonal are perfectly calibrated, and points below it show overconfidence. Under blank and mismatched images the curves drop further below the diagonal and ECE rises, which confirms that confidence does not track the loss of visual evidence. The effect appears for every model family and is worst on the harder PathVQA task and for MedGemma.

4.3 RQ4: Does confidence track graded corruption?

The blank and mismatch conditions take the visual evidence away all at once. To see whether confidence tracks a gradual loss of evidence, we degrade the image step by step for all three models, using two families of corruption: Gaussian blur (radius 1, 3, 5) and Gaussian noise ($\sigma=0.05, 0.1, 0.2$ of the pixel range). We then recompute accuracy and mean confidence at each level (Figure 2).

Answer. (RQ4) No, and the pattern holds across all three models. Blur lowers accuracy, most sharply for the medical model (MedGemma on VQA-RAD falls from 0.548 intact to 0.435 at blur radius 5, its ECE rising from 0.351 to 0.458), yet in every panel the confidence curve stays essentially flat and for Qwen even rises. Noise hurts accuracy less at these levels, again with no matching drop in confidence. Because confidence is anchored while accuracy erodes, the gap between them, and thus ECE, widens as the image degrades for every model and dataset.

4.4 RQ5: Is verbalized confidence any better?

Token-probability confidence is an internal number that a user never sees. A clinician would instead read whatever confidence the model states in words. So we run each model again with a prompt that asks it to add a line, “Confidence: $N\%$ ”, and read off the number. Table 3 compares the two signals.

Answer. (RQ5) No, and this holds for the two models that reliably comply. Qwen and MedGemma state a parseable confidence 94–100% of the time, and for both the verbalized number is higher and worse calibrated than the token probability in almost every condition; the starkest case is Qwen on VQA-RAD, which states more confidence on a blank gray image (0.893) than on the real one (0.883), driving verbalized ECE from 0.35 to 0.54. Asking for a confidence statement also inflates the token probability itself (Qwen’s VQA-RAD intact token ECE rises from 0.23 to 0.33 versus the plain prompt in Table 1). LLaVA-1.6, in contrast, emits a parseable confidence for only 13–26% of items, usually giving no readable number at all. So asking a medical VLM “how sure are you?”

Table 2: Closed (yes/no) versus open questions for all three models. In every case, closed-question ECE rises from the intact image to a mismatched one, while open questions stay overconfident throughout. Consistency (*Cons.*) under a mismatched image is high for the two 7B models but lower for MedGemma, which changes its answer more often yet is still overconfident. We show Intact and Mismatch.

Model	Dataset	Type	Acc $\uparrow$		ECE $\downarrow$		Cons.
			Intact	Mism.	Intact	Mism.	Mism.
Qwen2.5-VL-7B	VQA-RAD	Closed	0.753	0.625	0.047	0.212	0.622
		Open	0.305	0.205	0.468	0.563	0.310
	PathVQA	Closed	0.634	0.493	0.254	0.413	0.629
		Open	0.042	0.007	0.709	0.739	0.065
LLaVA-1.6-7B	VQA-RAD	Closed	0.629	0.530	0.219	0.322	0.645
		Open	0.200	0.130	0.572	0.635	0.380
	PathVQA	Closed	0.599	0.535	0.253	0.315	0.578
		Open	0.025	0.005	0.714	0.734	0.142
MedGemma-4B	VQA-RAD	Closed	0.665	0.454	0.226	0.425	0.283
		Open	0.400	0.190	0.507	0.707	0.155
	PathVQA	Closed	0.503	0.328	0.349	0.516	0.175
		Open	0.084	0.017	0.706	0.769	0.012

Table 3: Verbalized (Vrb, stated $N\%$) versus token-probability (Tok) ECE under a prompt that asks for a confidence score; lower is better. For the two models that reliably state a confidence (Qwen 94–100%, MedGemma $\sim 100\%$), verbalized confidence is worse calibrated than the token probability in almost every condition. [†]LLaVA-1.6 emitted a parseable confidence for only 13–26% of items, so its verbalized column covers a small, non-representative subset and should be read with caution.

Dataset	Condition	Qwen2.5-VL		LLaVA-1.6 [†]		MedGemma	
		Tok	Vrb	Tok	Vrb	Tok	Vrb
VQA-RAD	Intact	0.333	0.351	0.384	0.568	0.317	0.329
	Blank	0.474	0.542	0.450	0.401	0.580	0.523
	Mismatch	0.471	0.478	0.458	0.597	0.524	0.539
PathVQA	Intact	0.467	0.520	0.417	0.517	0.547	0.579
	Blank	0.562	0.596	0.502	0.371	0.681	0.702
	Mismatch	0.562	0.561	0.460	0.603	0.652	0.679

either makes it more overconfident or yields nothing usable; verbalized self-assessment is not a safe stand-in for a calibrated uncertainty estimate.

5 Discussion

Clinical implications. Leaderboards that report only accuracy hide the failure we describe. The mismatched-image case is the most dangerous: a wrong scan attached to the right patient, from a mislabeled study or a PACS error, yields a smooth, confident, wrong answer a clinician has little reason to doubt. Our closed-question result sharpens this: the yes/no questions that look best calibrated on the intact image are the very ones that collapse into a confident, image-independent guess once the image is corrupted. That medical pre-training does not help, and that the medical model is the most overconfident, warns against assuming a specialized model is safer.

Toward mitigation. The audit points to a few cheap safeguards. First, report calibration, not just accuracy: ECE and the high-confidence error rate under blank and mismatched images should go alongside any medical VQA accuracy claim. Second, do not trust verbalized confidence, since it is worse than the already poor token probability. Third, use simple consistency checks: because answers rarely change under a mismatched image (up to 0.63 for closed questions), testing whether the answer

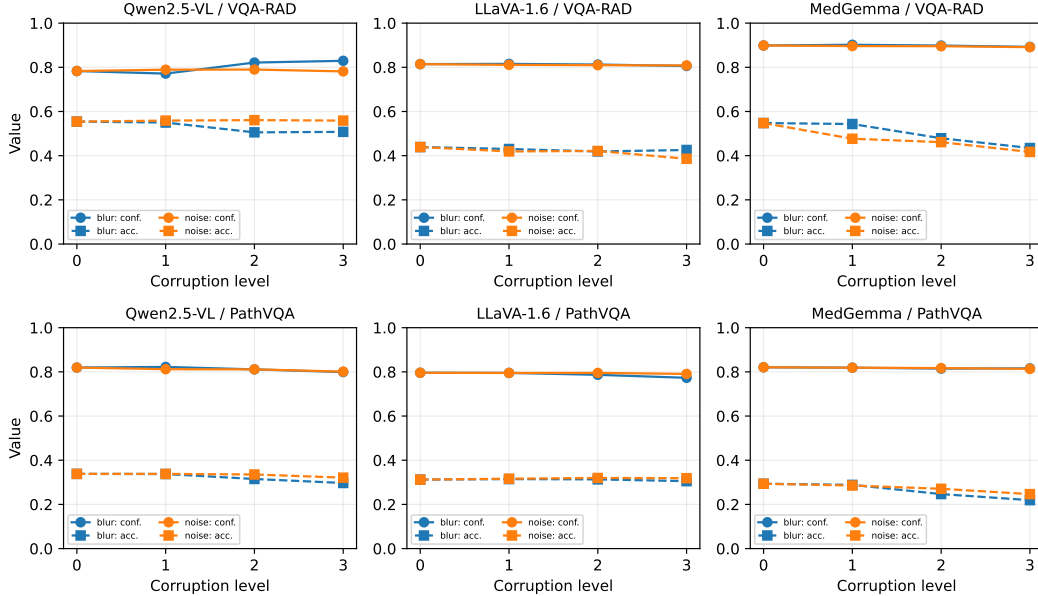


Figure 2: Accuracy (dashed, squares) and mean confidence (solid, circles) for all three models as corruption grows. The x-axis is the ordinal corruption level: 0 is the intact image, and levels 1–3 are blur radius 1/3/5 (blue) and noise $\sigma=0.05/0.1/0.2$ (orange). In every panel confidence stays flat while accuracy falls, so the confidence–accuracy gap, and thus ECE, widens; the effect is strongest for the medical model (MedGemma). A faithful model’s confidence curve would fall together with accuracy.

moves when the image is perturbed can flag responses that ignore it. We leave calibration-aware fine-tuning and post-hoc methods to future work.

Limitations. Our study has clear limits. It covers three locally run VLMs (two 7B general models and one 4B medical model); larger models, and other medically specialized ones such as LLaVA-Med, may calibrate differently. We use two datasets (radiology and pathology) and greedy decoding, so sampling-based confidence and other modalities remain untested. Our confidence score is a mean over generated tokens, which could be refined, for instance by averaging only over the answer span. Finally, ECE with 10 bins on small splits (VQA-RAD has only 451 items) can be noisy, so we report a bootstrap interval for the key closed-question result and otherwise stress effects that hold across both datasets and all three models.

6 Conclusion

We audited three locally run VLMs on radiology and pathology VQA, and we found a consistent problem: their confidence does not fall when the visual evidence they need is removed or replaced. Accuracy drops, yet confidence stays high, so calibration error rises sharply and many confident answers are wrong. The failure holds across model families and, tellingly, is worst for the medically specialized model. Most worrying for clinical use, the models react more to a blank image than to a believable but wrong one, which is exactly the case where a confidently incorrect answer is most likely to be trusted. The same overconfidence grows steadily as corruption increases, most sharply for the medical model, and the verbalized confidence a clinician would actually read is even worse calibrated than the internal signal for the models that state one at all. The failure is also concentrated in the closed questions that otherwise look most reliable. These failures are invisible to accuracy-only benchmarks, which is why we argue that calibration and image-drop checks should be a routine part of testing medical VLMs. We release our harness so these audits are easy to reproduce.

References

- Shuai Bai et al. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025. URL <https://arxiv.org/abs/2502.13923>. Checkpoint: <https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct>; Apache-2.0.
- Zijie Cheng, Ariel Yuhan Ong, Siegfried K. Wagner, et al. Understanding the robustness of vision-language models to medical image artefacts. *npj Digital Medicine*, 8(1):727, 2025. doi: 10.1038/s41746-025-02108-w.
- Yuetian Du, Yucheng Wang, Ming Kong, Tian Liang, Qiang Long, Bingdi Chen, and Qiang Zhu. Confidence calibration for multimodal LLMs: An empirical study through medical VQA. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*, volume 15965 of *Lecture Notes in Computer Science*, pages 89–99, 2025. doi: 10.1007/978-3-032-04978-0_9.
- Zishan Gu, Jiayuan Chen, Fenglin Liu, Changchang Yin, and Ping Zhang. MedVH: Toward systematic evaluation of hallucination for large vision language models in the medical context. *Advanced Intelligent Systems*, 8(1):e202500255, 2026. doi: 10.1002/aisy.202500255.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017. URL <https://proceedings.mlr.press/v70/guo17a.html>.
- Xuehai He, Zhuo Cai, Wenlan Wei, Yichen Zhang, Luntian Mou, Eric Xing, and Pengtao Xie. Towards visual question answering on pathology images. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pages 708–718, 2021. doi: 10.18653/v1/2021.acl-short.90. URL <https://aclanthology.org/2021.acl-short.90/>. Dataset mirror: <https://huggingface.co/datasets/flaviagiammarino/path-vqa>; MIT.
- Beria Chingnabe Kalpelbe, Angel Gabriel Adaambiik, and Wei Peng. Vision language models in medicine. *arXiv preprint arXiv:2503.01863*, 2025. URL <https://arxiv.org/abs/2503.01863>.
- Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://openreview.net/forum?id=VD-AYtP0dve>.
- Jason J. Lau, Soumya Gayen, Asma Ben Abacha, and Dina Demner-Fushman. A dataset of clinically generated visual questions and answers about radiology images. *Scientific Data*, 5(1):180251, 2018. doi: 10.1038/sdata.2018.251. Dataset mirror: <https://huggingface.co/datasets/flaviagiammarino/vqa-rad>; CC0-1.0.
- Khai Le-Duc et al. S-Chain: Structured visual chain-of-thought for medicine. *arXiv preprint arXiv:2510.22728*, 2025. URL <https://arxiv.org/abs/2510.22728>.
- Zehui Liao, Shishuai Hu, Ke Zou, Huazhu Fu, Liangli Zhen, and Yong Xia. Vision-amplified semantic entropy for hallucination detection in medical visual question answering. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*, volume 15964 of *Lecture Notes in Computer Science*, pages 669–679, 2025. doi: 10.1007/978-3-032-04971-1_63.
- Zehui Liao, Shishuai Hu, Ke Zou, Mengyuan Jin, Yanning Zhang, Huazhu Fu, Liangli Zhen, and Yong Xia. UniVRSE: Unified vision-conditioned response semantic entropy for hallucination detection in medical vision-language models. *arXiv preprint arXiv:2503.20504*, 2026. URL <https://arxiv.org/abs/2503.20504>. Version 2; expanded and retitled from the VASE preprint.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in words. *Transactions on Machine Learning Research*, 2022. URL <https://openreview.net/forum?id=8s8K2UZGTZ>.
- Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. LLaVA-NeXT: Improved reasoning, OCR, and world knowledge. LLaVA project blog, 2024. URL <https://llava-vl.github.io/blog/2024-01-30-llava-next/>. Checkpoint: <https://huggingface.co/llava-hf/llava-v1.6-mistral-7b-hf>; Apache-2.0.

- Johannes Moll, Markus Graf, Tristan Lemke, Nicolas Lenhart, Daniel Truhn, Jean-Benoit Delbrouck, Jiazhen Pan, Daniel Rueckert, Lisa C. Adams, and Keno K. Bressem. Evaluating reasoning faithfulness in medical vision-language models using multimodal perturbations. *arXiv preprint arXiv:2510.11196*, 2025. URL <https://arxiv.org/abs/2510.11196>.
- Sofiane Ouaari, Kevin Vorwalder, and Nico Pfeifer. Assessing VLM reliability for medical image quality evaluation under corruption and bias. *arXiv preprint arXiv:2607.01973*, 2026. URL <https://arxiv.org/abs/2607.01973>.
- Andrew Sellergren et al. MedGemma technical report. *arXiv preprint arXiv:2507.05201*, 2025. URL <https://arxiv.org/abs/2507.05201>. Checkpoint: <https://huggingface.co/google/medgemma-4b-it>; Health AI Developer Foundations terms.
- Eren Senoglu, Federico Toschi, Nicolo Brunello, Andrea Sassella, and Mark James Carman. Just how sure are you? improving verbalized uncertainty calibration in medical VQA. *arXiv preprint arXiv:2606.27023*, 2026. URL <https://arxiv.org/abs/2606.27023>.
- Jinge Wu, Yunsoo Kim, and Honghan Wu. Hallucination benchmark in medical visual question answering. In *International Conference on Learning Representations (ICLR), Tiny Papers Track*, 2024. URL <https://openreview.net/forum?id=vxlXq0j4zv>.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Introduction state the central claim—that confidence fails to fall when visual evidence is removed, replaced, or corrupted, so calibration error rises—and the Results (RQ1–RQ5) and Tables 1–3 support exactly these claims, with scope limited to the three evaluated models and two datasets.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The Discussion contains a dedicated Limitations paragraph covering the three-model/two-dataset scope, greedy decoding, the mean-over-tokens confidence proxy, and small-split ECE noise (for which a bootstrap interval is reported).

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper is an empirical audit and contains no theoretical results or proofs.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The Setup describes the exact models, datasets, image conditions, decoding settings (greedy, at most 32 new tokens), confidence definition, and ECE binning; the evaluation harness is released, and all datasets and model checkpoints are public.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: We release the evaluation harness (model-agnostic inference, scoring, stratification, bootstrap CIs, and plotting scripts) with usage instructions; both datasets (VQA-RAD, PathVQA) and all three model checkpoints are publicly available on the Hugging Face Hub as identified in the bibliography.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: No training is performed; the Setup specifies the test splits (VQA-RAD $n=451$, PathVQA $n=6719$), the three image conditions, greedy decoding with at most 32 new tokens, and the 10-bin ECE used throughout.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: For the key RQ3 result we report a bootstrap 95% confidence interval (5000 resamples of items with replacement, percentile interval) on the intact-to-mismatch ECE increase for closed questions; the interval excludes zero. Other metrics are computed on the full test splits.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All experiments are inference only (no training). Each model/dataset run used a single NVIDIA A100 GPU; the heaviest runs (the full severity sweep over PathVQA, seven conditions $\times$ 6719 items) take on the order of hours, and total compute is modest.

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work uses publicly available, de-identified medical VQA datasets and performs inference only; it involves no human subjects and is aimed at improving the safety and reliability of medical VLMs.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The Discussion (Clinical implications) describes the positive impact—surfacing a safety-relevant calibration failure invisible to accuracy-only benchmarks and recommending cheap safeguards—and the motivating negative scenario of confidently wrong answers on degraded or mismatched medical images.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: We release only an evaluation harness (analysis code); no new model, generator, or scraped dataset with misuse risk is released.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All datasets and models are cited, and their licenses/terms are noted in the bibliography (VQA-RAD CC0-1.0, PathVQA MIT, Qwen2.5-VL and LLaVA-1.6 Apache-2.0, MedGemma under the Health AI Developer Foundations terms).

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The only new asset is the evaluation harness, which is released with a README describing dependencies, the datasets/models to download, and exact commands to reproduce each experiment.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing or research with human subjects; it uses existing public datasets.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The paper does not involve human subjects, so no IRB approval is required.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: The vision-language models in this paper are the objects of evaluation, not tools used to develop the methodology; no LLM is an important, original, or non-standard component of the core method.